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Gap inspection apparatus and gap inspection method

Abstract

A gap inspection apparatus and a gap inspection method are disclosed. The gap inspection apparatus is configured for a wound battery cell, where in a height direction of the wound battery cell. The gap inspection apparatus includes: an X-ray source and an X-ray detector and a bearing mechanism located between the X-ray source and the X-ray detector. The bearing mechanism includes at least one accommodating groove, the accommodating groove is configured to accommodate the wound battery cell, and the height direction of the wound battery cell crosses a first direction in which the X-ray source points to the X-ray detector. The X-ray source is configured to irradiate an arc-shaped area of the wound body portion, and the X-ray detector is configured to obtain an imaging image of the irradiated arc-shaped area, so as to obtain a gap value between two electrode plates within the arc-shaped area.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International application PCT/CN2023/085237 filed on Mar. 30, 2023 that claims priority to Chinese Patent Application No. 202211290267.X filed Oct. 20, 2022 which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) This application relates to the field of battery cell inspection technologies, and specifically, to a gap inspection apparatus and a gap inspection method.

BACKGROUND

(2) With rapid development of the new energy industry, electric vehicles that use lithium-ion batteries as traction batteries have begun to be widely used. Quality of the traction battery not only affects service life of the traction battery, but also affects safety of driving.

(3) The traction battery includes several battery cells, and during cycling of the traction battery, a gap value between a positive electrode plate and a negative electrode plate of the battery cell is an important factor affecting migration of lithium ions. For example, when the gap value is large, lithium-ion accumulation and lithium precipitation are likely to occur, resulting in safety risks. Therefore, the gap value needs to be accurately measured during production of the traction battery.

SUMMARY

(4) This application provides a gap inspection apparatus and a gap inspection method, so as to measure a gap value between two electrode plates of a wound battery cell in a simple and efficient way.

(5) To resolve the foregoing technical problems, one technical solution used in this application is implemented as follows: providing a gap inspection apparatus, configured for a wound battery cell, where in a height direction of the wound battery cell, the wound battery cell includes a wound body portion and a tab that are connected to each other, and the gap inspection apparatus includes: an X-ray source and an X-ray detector that are oppositely disposed, and a bearing mechanism located between the X-ray source and the X-ray detector, where the bearing mechanism includes at least one accommodating groove, the accommodating groove is configured to accommodate the wound battery cell, and the height direction of the wound battery cell crosses a first direction in which the X-ray source points to the X-ray detector; and the X-ray source is configured to irradiate an arc-shaped area of the wound body portion, and the X-ray detector is configured to obtain an imaging image of the irradiated arc-shaped area, so as to obtain a gap value between two electrode plates within the arc-shaped area based on the imaging image.

(6) With the foregoing gap inspection apparatus, multi-angle image acquisition, three-dimensional reconstruction, CT tomogram, and other processes are not required for the wound battery cell, and the imaging image of the arc-shaped area of the wound battery cell can be directly obtained through the X-ray source and the X-ray detector, so as to obtain the gap value between the two electrode plates within the arc-shaped area. The inspection process is simple and efficient.

(7) In some embodiments, the bearing mechanism includes a bearing member and a cushion fastener, an accommodating cavity is included on a side of the bearing member facing the X-ray

source or the X-ray detector, the cushion fastener is located in the accommodating cavity, and the accommodating groove is disposed on the cushion fastener. The foregoing bearing mechanism 4 has a simple structure and is easy to manufacture.

(8) In some embodiments, in the first direction, the accommodating groove runs through the cushion fastener. Such design can reduce an impact of the cushion fastener on an imaging effect of the wound battery cell.

(9) And/or, a Shore hardness of the cushion fastener is greater than or equal to 40 degrees and less than or equal to 50 degrees. Such design allows the cushion fastener to have both a position limiting effect and a damage reduction effect.

(10) And/or, the cushion fastener includes foam. The cushion fastener made of this material has advantages of good X-ray penetration, light weight, and the like to reduce the impact of the cushion fastener on the imaging effect of the wound battery cell.

(11) In some embodiments, the accommodating groove includes a first accommodating sub-groove and a second accommodating sub-groove that communicate with each other, the first accommodating sub-groove is configured to accommodate the wound body portion, and the second accommodating sub-groove is configured to accommodate the tab, where a preset distance is present between the wound body portion and an inner wall of the first accommodating sub-groove, and a preset distance is present between the tab and an inner wall of the second accommodating sub-groove. Such design can reduce damage to the wound battery cell by the accommodating groove.

(12) In some embodiments, two tabs are provided and arranged on a same side of the wound body portion; and the second accommodating sub-groove is configured to accommodate the two tabs. Such design reduces the difficulty of alignment during placement.

(13) In some embodiments, in a second direction, one pick-and-place slot is disposed on either of two opposite sides of the first accommodating sub-groove, where the second direction is perpendicular to a third direction in which the first accommodating sub-groove points to the second accommodating sub-groove, and a size of the accommodating groove in the third direction is larger than that of the pick-and-place slot. The introduction of the pick-and-place slot can reduce difficulty of picking up or placing the wound battery cell, and reduce the damage to the wound battery cell.

(14) In some embodiments, the cushion fastener is provided with a first error-proofing portion, a second error-proofing portion is disposed in the accommodating cavity, and the first error-proofing portion and the second error-proofing portion cooperate with each other. Such design can reduce the probability of placing the cushion fastener in the bearing member in a wrong direction, improving inspection efficiency.

(15) In some embodiments, the bearing member includes a base plate and a side plate that are connected to each other, and the side plate is arranged around the base plate to form the accommodating cavity, where a material of the base plate includes carbon fiber, and a material of the side plate includes metal. The base plate made of the foregoing material can reduce the impact on the imaging effect of the wound battery cell. The side plate made of the foregoing material can enable the base plate to have a specific bearing capacity, and the side plate limits the cushion fastener, which helps improve a position limiting effect of the cushion fastener on the wound battery cell.

(16) In some embodiments, the bearing member further includes two handles that are oppositely disposed and located on an outer surface of a side of the side plate facing away from the accommodating cavity. The design of the handles allows for easy transport.

(17) To resolve the foregoing technical problems, another technical solution used in this application is implemented as follows: providing a gap inspection method that uses the gap inspection apparatus according to any one of the foregoing embodiments. The gap inspection method includes: placing a wound battery cell in an accommodating groove of a bearing mechanism, and

disposing the bearing mechanism between an X-ray source and an X-ray detector that are oppositely disposed, where in a height direction of the wound battery cell, the wound battery cell includes a wound body portion and a tab that are connected to each other, and the height direction of the wound battery cell crosses a first direction in which the X-ray source points to the X-ray detector; making the X-ray source irradiate an arc-shaped area of the wound body portion, so that the X-ray detector obtains an imaging image of the arc-shaped area; and obtaining, based on the imaging image, a gap value between two electrode plates within the arc-shaped area.

(18) In some embodiments, the wound body portion includes a positive electrode plate and a negative electrode plate that are alternately arranged; and the step of obtaining, based on the imaging image, a gap value between two electrode plates in the wound body portion within the arc-shaped area includes: pre-processing the imaging image and obtaining a target area from the pre-processed imaging image; determining position information of the positive electrode plate and the negative electrode plate from the target area based on size information of the positive electrode plate and the negative electrode plate; and obtaining, from the target area based on the position information of the positive electrode plate and the negative electrode plate, a gap value between the positive electrode plate and the negative electrode plate that are adjacent to each other.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) To describe the technical solutions in the embodiments of this application more clearly, the following briefly describes the accompanying drawings required for describing the embodiments. Apparently, the accompanying drawings in the following descriptions show merely some embodiments of this application, and persons of ordinary skill in the art may still derive other drawings from the accompanying drawings without creative efforts.

(2) FIG. 1 is a schematic structural diagram of a wound battery cell according to an embodiment;

(3) FIG. 2 is a schematic cross-sectional view of the wound battery cell in FIG. 1 along line A-A according to an embodiment;

(4) FIG. 3 is a schematic structural diagram of a wound battery cell according to another embodiment;

(5) FIG. 4 is a schematic structural diagram of a gap inspection apparatus according to an embodiment of this application;

(6) FIG. 5 is a schematic exploded view of a bearing mechanism in FIG. 4 according to an embodiment;

(7) FIG. 6 is a schematic structural diagram of a cushion fastener in FIG. 4 according to another embodiment;

(8) FIG. 7 is a schematic structural diagram of a cushion fastener in FIG. 4 according to another embodiment;

(9) FIG. 8 is a schematic flowchart of a gap inspection method according to an embodiment of this application;

(10) FIG. 9 is a schematic flowchart of step S103 in FIG. 8 corresponding to an embodiment; and

(11) FIG. 10 is a schematic diagram of a target area obtained after step S201 in FIG. 9 according to an embodiment.

(12) Specific reference signs are as follows: wound battery cell **1**, wound body portion **10**, tab **12**; positive electrode plate **100**, negative electrode plate **102**, separator **104**, positive electrode tab **120**, negative electrode tab **122**, arc-shaped area **106**, planar area **108**; positive electrode current collector **1000**, positive electrode active substance layer **1002**, negative electrode current collector **1020**, negative electrode active substance layer **1022**; gap inspection apparatus **2**, X-ray source **20**, X-ray detector **22**, bearing mechanism **24**; accommodating groove **240**, bearing member **242**,

cushion fastener **244**, accommodating cavity **246**, pick-and-place slot **248**, first error-proofing portion **241**, second error-proofing portion **243**; base plate **2420**, side plate **2422**, handle **2424**, first accommodating sub-groove **2400**, second accommodating sub-groove **2402**; height direction P1 of the wound battery cell **1**, width direction P2 of the wound battery cell **1**, thickness direction P3 of the wound battery cell **1**, first direction L1, second direction L2, and third direction L3.

DESCRIPTION OF EMBODIMENTS

(13) The following clearly and completely describes the technical solutions in the embodiments of this application with reference to the accompanying drawings in the embodiments of this application. Apparently, the described embodiments are only some but not all of the embodiments of this application. All other embodiments obtained by persons of ordinary skill in the art based on the embodiments of this application without creative efforts shall fall within the protection scope of this application.

(14) Unless otherwise defined, all technical and scientific terms used herein shall have the same meanings as commonly understood by those skilled in the art to which this application relates. The terms used herein are intended to merely describe the specific embodiments rather than to limit this application. The terms “include”, “comprise”, and “have” and any other variations thereof in the specification, claims and brief description of drawings of this application are intended to cover non-exclusive inclusions.

(15) In the descriptions of the embodiments of this application, the terms “first”, “second”, and the like are merely intended to distinguish between different objects, and shall not be understood as any indication or implication of relative importance or any implicit indication of the number, specific sequence, or primary-secondary relationship of the technical features indicated. In the descriptions of the embodiments of this application, “a plurality of” means more than two, unless otherwise specifically defined.

(16) In this specification, reference to “embodiment” means that specific features, structures, or characteristics described with reference to the embodiment may be incorporated in at least one embodiment of this application. The word “embodiment” appearing in various places in the specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. It is explicitly or implicitly understood by persons skilled in the art that the embodiments described herein may be combined with other embodiments.

(17) In the descriptions of the embodiments of this application, the term “and/or” is only an associative relationship for describing associated objects, indicating that three relationships may be present. For example, A and/or B may indicate the following three cases: presence of only A, presence of both A and B, and presence of only B. In addition, the character “/” in this specification generally indicates an “or” relationship between contextually associated objects.

(18) In the descriptions of the embodiments of this application, the term “a plurality of” means two (inclusive) or more. Similarly, “a plurality of groups” means two (inclusive) groups or more, and “a plurality of pieces” means two (inclusive) pieces or more.

(19) In the descriptions of the embodiments of this application, the orientations or positional relationships indicated by the technical terms “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “perpendicular”, “horizontal”, “top”, “bottom”, “inside”, “outside”, “clockwise”, “counterclockwise”, “axial”, “radial”, “circumferential”, and the like are based on the orientations or positional relationships shown in the accompanying drawings. These terms are merely for ease and brevity of description of the embodiments of this application rather than indicating or implying that the apparatuses or elements mentioned must have specific orientations or must be constructed or manipulated according to specific orientations, and therefore shall not be construed as any limitation on the embodiments of this application.

(20) In the descriptions of the embodiments of this application, unless otherwise specified and

defined explicitly, the technical terms “mount”, “connect”, “join”, and “fasten” should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, or an interaction between two elements. Persons of ordinary skill in the art can understand specific meanings of these terms in the embodiments of this application as appropriate to specific situations.

(21) Currently, from a perspective of market development, application of traction batteries is becoming more and more extensive. Traction batteries have been not only used in energy storage power supply systems such as hydroelectric power plants, thermal power plants, wind power plants, and solar power plants, but also widely used in many other fields including electric transportation tools such as electric bicycles, electric motorcycles, and electric vehicles, military equipment, and aerospace. With continuous expansion of application fields of traction batteries, market demands for the traction batteries are also increasing, and the traction batteries including wound battery cells are particularly widely used.

(22) Refer to FIG. 1 and FIG. 2. FIG. 1 is a schematic structural diagram of a wound battery cell according to an embodiment. FIG. 2 is a schematic cross-sectional view of the wound battery cell in FIG. 1 along line A-A according to an embodiment. First, it is defined that a wound battery cell **1** includes a height direction **P1**, a width direction **P2**, and a thickness direction **P3** that are perpendicular to each other. In the height direction **P1** of the wound battery cell **1**, the wound battery cell **1** includes a wound body portion **10** and a tab **12** that are connected to each other. The wound body portion **10** is generally formed by winding a positive electrode plate **100** and a negative electrode plate **102**, and a separator **104** is disposed between the positive electrode plate **100** and the negative electrode plate **102**. The positive electrode plate **100** includes a positive electrode current collector **1000** and a positive electrode active substance layer **1002** applied on a surface of the positive electrode current collector **1000**, and the negative electrode plate **102** includes a negative electrode current collector **1020** and a negative electrode active substance layer **1022** applied on a surface of the negative electrode current collector **1020**. In this case, the corresponding tab **12** includes a positive electrode tab **120** and a negative electrode tab **122**. The positive electrode tab **120** is connected to the positive electrode plate **100**, and the negative electrode tab **122** is connected to the negative electrode plate **102**. As shown in FIG. 1, the positive electrode tab **120** and the negative electrode tab **122** may be located on a same side of the wound body portion **10**. Certainly, in another embodiment, the positive electrode tab **120** and the negative electrode tab **122** may alternatively be respectively located on opposite sides of the wound body portion **10**. This is not limited in this application. In addition, as shown in FIG. 1, the wound body portion **10** of the wound battery cell **1** may be square. Certainly, in another embodiment, as shown in FIG. 3, the wound body portion **10** of the wound battery cell **1** may alternatively be cylindrical. This is not limited in this application.

(23) Further, as shown in FIG. 1, FIG. 2, and FIG. 3, the wound body portion **10** includes an arc-shaped area **106**, and the arc-shaped area **106** may be any area with curvature (that is, a non-planar area) on the wound body portion **10**, for example, a left or right corner of the wound body portion **10** in FIG. 1; and for another example, any part of the wound body portion **10** in FIG. 2.

(24) The inventors of this application have noticed that, for the wound battery cell **1**, during winding of the wound battery cell **1**, due to difference between a radius of an inner circle and a radius of an outer circle of the arc-shaped area **106**, forces applied on the inner circle and the outer circle are different, resulting in an excessively large gap value may between the negative electrode plate **102** and the positive electrode plate **100** of the inner circle after hot pressing. Currently, a gap value test for the wound battery cell **1** generally adopts the following CT slice inspection method: performing multi-angle image acquisition on the wound battery cell **1**, performing three-dimensional reconstruction based on a plurality of acquired images, and then intercepting a model

obtained by the three-dimensional reconstruction to obtain a CT tomogram in a preset direction and obtain a corresponding gap value based on the CT tomogram. Obviously, the inspection method has problems of low inspection efficiency and high inspection cost.

(25) To resolve the foregoing technical problems, refer to FIG. 4. FIG. 4 is a schematic structural diagram of a gap inspection apparatus according to an embodiment of this application. The gap inspection apparatus 2 is configured to measure the gap value of the wound battery cell 1, and includes an X-ray source 20, an X-ray detector 22, and a bearing mechanism 24. The X-ray source 20 and the X-ray detector 22 are arranged opposite to each other and spaced apart, and the bearing mechanism 24 is located between the X-ray source 20 and the X-ray detector 22. The bearing mechanism 24 includes at least one accommodating groove 240, the accommodating groove 240 is configured to accommodate the wound battery cell 1, and during the inspection, the height direction P1 of the wound battery cell 1 crosses a first direction L1 in which the X-ray source 20 points to the X-ray detector 22; and the X-ray source 20 is configured to irradiate the arc-shaped area 106 (as shown in FIG. 1, FIG. 2, and FIG. 3) of the wound body portion 10, and the X-ray detector 22 is configured to obtain an imaging image of the irradiated arc-shaped area 106, so as to obtain a gap value between two electrode plates within the arc-shaped area 106 based on the imaging image.

(26) Specifically, the X-ray source 20 is an apparatus capable of emitting an X-ray. The X-ray detector 22 may be a flat panel detector capable of converting received X-ray energy into a recordable electrical signal by measuring the X-ray energy received by the X-ray detector 22 and generating an electrical signal proportional to an X-ray to form a corresponding image. Optionally, the X-ray detector 22 may include three parts: an X-ray conversion module, a photoelectric conversion module, and a signal reading and transmission module. During the test of the gap inspection apparatus 2, an electron emitted from a cathode of the X-ray source 20, after being accelerated by an electric field between the cathode and an anode, bombards an X-ray source target, to transfer kinetic energy of the electron to an atom on the target. About 1% of the energy is converted into an X-ray, and the X-ray is emitted from an X-ray irradiation window. The emitted X-ray penetrates the wound battery cell 1, and then is imaged by the X-ray detector 22.

(27) In an application scenario, as shown in FIG. 2, an inspection resolution for a gap value d between the negative electrode plate 102 and the positive electrode plate 100 that are adjacent to each other in the wound battery cell 1 needs to be less than 20 microns. To satisfy this inspection requirement, on the one hand, the X-ray source 20 in this application selects an integral microfocus X-ray source. Generally, a smaller focal spot size of the X-ray source 20 indicates a higher imaging image resolution and a clearer imaging picture. The integral microfocus X-ray source has advantages of high stability, low repair rate, small size, easy operation and installation, applicability for a production line, and the like. In this application, to improve X-ray transmittance and improve a signal-to-noise ratio of the imaging image of the wound battery cell 1, a 150 KV microfocus X-ray source is used. On the other hand, in this application, a selected pixel size of the X-ray detector 22 divided by a magnification needs to be less than 20 microns. Optionally, when the magnification is greater than or equal to $5\times$, the pixel size of the X-ray detector 22 is less than or equal to 100 microns.

(28) The height direction P1 of the wound battery cell 1 crossing a first direction L1 in which the X-ray source 20 points to the X-ray detector 22 indicates that the wound battery cell 1 is lying between the X-ray source 20 and the X-ray detector 22. Preferably, the height direction P1 of the wound battery cell 1 and the first direction L1 are perpendicular to each other, and the first direction L1 and a thickness direction P3 of the wound battery cell 1 are parallel to each other. The X-ray emitted by the X-ray source 20 can penetrate the wound battery cell 1 from the thickness direction P3 of the wound battery cell 1.

(29) Refer to FIG. 2 again. The arc-shaped area 106 may include any area with curvature (that is, a non-planar area) on the wound body portion 10. In the thickness direction P3, the positive electrode

plate **100** and the negative electrode plate **102** that are adjacent to each other within the arc-shaped area **106** are not completely blocked from each other. After irradiation by the X-ray source **20**, the X-ray detector **22** is able to obtain an imaging image of the positive electrode plate **100** and the negative electrode plate **102** that are spaced apart within the arc-shaped area **106**. In addition, as shown in FIG. **1** and FIG. **2**, when the wound body portion **10** further includes a planar area **108**, the positive electrode plate **100** and the negative electrode plate **102** that are adjacent to each other are completely blocked from each other in the thickness direction P3. After irradiation by the X-ray source **20**, in the imaging image obtained by the X-ray detector **22**, the positive electrode plate **100** and the negative electrode plate **102** that are adjacent to each other overlap, and the gap value d between the positive electrode plate **100** and the negative electrode plate **102** cannot be accurately obtained. That is, the X-ray source **20** in the gap inspection apparatus **2** provided in this application needs to irradiate to the arc-shaped area **106** of the wound body portion **10**.

(30) In addition, it can be known that after the X-ray detector **22** obtains an imaging image of the irradiated arc-shaped area **106**, a gap value between any two electrode plates within the arc-shaped area **106** can be obtained based on the imaging image. The any two electrode plates may be the positive electrode plate **100** and the negative electrode plate **102** that are adjacent to each other, two adjacent positive electrode plates **100** that are spaced apart, or two adjacent negative electrode plates **102** that are spaced apart. In addition, the any two electrode plates may be located on an innermost side of the arc-shaped area **106** or at other positions of the arc-shaped area **106**. This is not limited in this application.

(31) In summary, with the foregoing gap inspection apparatus **2**, multi-angle image acquisition, three-dimensional reconstruction, CT tomogram, and other processes are not required for the wound battery cell **1**, and the imaging image of the arc-shaped area **106** of the wound battery cell **1** can be directly obtained through the X-ray source **20** and the X-ray detector **22**, so as to obtain the gap value between the two electrode plates within the arc-shaped area **106**. The inspection process is simple and efficient.

(32) Refer to FIG. **5**. FIG. **5** is a schematic exploded view of a bearing mechanism in FIG. **4** according to an embodiment. The bearing mechanism **24** includes a bearing member **242** and a cushion fastener **244**, an accommodating cavity **246** is included on a side of the bearing member **242** facing the X-ray source **20** or the X-ray detector **22**, the cushion fastener **244** is located in the accommodating cavity **246**, and the accommodating groove **240** is disposed on the cushion fastener **244**. The foregoing bearing mechanism **24** has a simple structure and is easy to manufacture.

(33) Specifically, the accommodating cavity **246** being included on a side of the bearing member **242** facing the X-ray source **20** or the X-ray detector **22** indicates that when the X-ray source **20** is located above the bearing mechanism **24** and the X-ray detector **22** is located below the bearing mechanism **24**, the accommodating cavity **246** is included on a side of the bearing member **242** facing the X-ray source **20** above the bearing member **242**. When the X-ray detector **22** is located above the bearing mechanism **24** and the X-ray source **20** is located below the bearing mechanism **24**, the accommodating cavity **246** is included on a side of the bearing member **242** facing the X-ray detector **22** above the bearing member **242**.

(34) The accommodating cavity **246** is a non-closed cavity, and the accommodating cavity **246** has an opening on a side facing the X-ray source **20** or the X-ray detector **22**. On the one hand, the cushion fastener **244** and the wound battery cell **1** under test can easily enter and exit the bearing member **242** from the opening; on the other hand, a blocking effect on the X-ray from the bearing member **242** can be reduced to improve a subsequent imaging effect. When the side of the bearing member **242** facing the X-ray source **20** includes the accommodating cavity **246**, the X-ray emitted by the X-ray source **20** sequentially penetrates the wound battery cell **1** and the bearing member **242** on a side of the wound battery cell **1**. When the side of the bearing member **242** facing the X-ray detector **22** includes the accommodating cavity **246**, the X-ray emitted by the X-ray source sequentially penetrates the bearing member **242** on a side of the wound battery cell **1** and the

wound battery cell **1**.

(35) As shown in FIG. 5, the bearing member **242** includes a base plate **2420** and a side plate **2422** that are connected to each other, and the side plate **2422** is arranged around the base plate **2420** to form the accommodating cavity **246**, that is, in this case, the side plate **2422** is of an annular structure. The bearing member **242** has a simple structural design and is easy to manufacture.

(36) Optionally, a material of the base plate **2420** includes carbon fiber. In this case, the base plate **2420** is a carbon fiber plate formed by infiltrating and hardening carbon fibers arranged in a same direction with resin, which has advantages of low density, good X-ray penetration, light weight, good flexibility, good durability, corrosion resistance, and the like, that is, the base plate **2420** made of this material can reduce an impact on the imaging effect of the wound battery cell **1**. Preferably, a thickness of the base plate **2420** in the first direction **L1** is 1.5 mm to 2.5 mm (for example, 2 mm).

(37) Optionally, a material of the side plate **2422** includes metal (for example, stainless steel). The side plate **2422** does not impact the imaging effect of the wound battery cell **1**, and therefore the side plate **2422** does not require high X-ray penetration. The side plate **2422** made of the foregoing metal material is able to fasten the base plate **2420**, so that the base plate **2420** has a specific bearing capacity, and the side plate **2422** limits the cushion fastener **244**, which helps improve a position limiting effect of the cushion fastener **244** on the wound battery cell **1**, thereby further improving accuracy of a testing position.

(38) Optionally, the bearing member **242** further includes two handles **2424** that are oppositely disposed and located on an outer surface of a side of the side plate **2422** facing away from the accommodating cavity **246**. The design of the handles **2424** allows for easy transport. Preferably, a material of the handle **2424** is metal, and the handle **2424** may be integrally formed with the side plate **2422**. In addition, a shape of the handle **2424** may be arc-shaped and the like, to facilitate grasping.

(39) The cushion fastener **244** is a member that can be deformed to a specific extent and can limit and fasten the wound battery cell **1**. On the one hand, when the device shakes or moves, the cushion fastener **244** can cushion a stress by deformation of the cushion fastener **244**, reducing the probability that the position of the wound battery cell **1** deviates. On the other hand, during fetching of the wound battery cell **1**, the cushion fastener **244** has little interference with the wound battery cell **1**, which can reduce damage to the wound battery cell **1**.

(40) Optionally, the cushion fastener **244** includes foam. For example, the cushion fastener **244** is made of EVA (ethylene-vinyl acetate copolymer) and the like. The cushion fastener **244** made of this material has advantages of good X-ray penetration, light weight, and the like to reduce the impact of the cushion fastener **244** on the imaging effect of the wound battery cell **1**.

(41) Optionally, a Shore hardness of the cushion fastener **244** is greater than or equal to 40 degrees and less than or equal to 50 degrees. For example, the Shore hardness of the cushion fastener **244** is 45 degrees or the like. Generally, a lower hardness indicates a greater deformation capacity of the cushion fastener **244** and a poorer position limiting effect on the wound battery cell **1**. A higher hardness indicates a greater probability of the cushion fastener **244** damaging the wound battery cell **1** when the wound battery cell **1** is picked up or placed. Therefore, the cushion fastener **244** in the foregoing Shore hardness range is designed based on comprehensive consideration of the two factors, so that the cushion fastener **244** can have both a position limiting effect and a damage reduction effect.

(42) Optionally, in the first direction **L1**, the accommodating groove **240** runs through the cushion fastener **244**. Specifically, the base plate **2420** is exposed from the accommodating groove **240**. Compared to the accommodating groove **240** not running through the cushion fastener **244**, in such design, the X-ray needs to penetrate only the arc-shaped area **106** of the wound battery cell **1** and the base plate **2420** at a corresponding position before being received by the X-ray detector **22**, to reduce the impact of the cushion fastener **244** on the imaging effect of the wound battery cell **1**.

(43) Still refer to FIG. 4 and FIG. 5. The accommodating groove **240** includes a first accommodating sub-groove **2400** and a second accommodating sub-groove **2402** that communicate with each other. The first accommodating sub-groove **2400** is configured to accommodate the wound body portion **10**, and the second accommodating sub-groove **2402** is configured to accommodate the tab **12**. A preset distance is present between the wound body portion **10** and an inner wall of the first accommodating sub-groove **2400**, and a preset distance is present between the tab **12** and an inner wall of the second accommodating sub-groove **2402**. In other words, a size of the accommodating groove **240** is larger than a size of the wound battery cell **1** to be accommodated in the accommodating groove **240**, so as to reduce damage to the wound battery cell **1** by the accommodating groove **240**.

(44) Optionally, the preset distance is 0.5 mm to 1.5 mm (for example, 1 mm). In such design, the size of the accommodating groove **240** is slightly larger than the size of the wound battery cell **1** be accommodated in the accommodating groove **240**, so as to reduce the damage to the wound battery cell **1** by the accommodating groove **240** and ensure the position limiting effect of the cushion fastener **244** on the wound battery cell **1**.

(45) Optionally, as shown in FIG. 4, a second direction **L2** and a third direction **L3** that are perpendicular to each other are defined. A direction in which the first accommodating sub-groove **2400** points to the second accommodating sub-groove **2402** is the third direction **L3**. The third direction **L3** and the height direction **P1** of the wound battery cell **1** are parallel to each other, and the second direction **L2** and the width direction **P2** of the wound battery cell **1** are parallel to each other. Generally, in the second direction **L2**, a width of the wound body portion **10** is greater than a width of the tab **12**. Correspondingly, a width of the first accommodating sub-groove **2400** is greater than a width of the second accommodating sub-groove **2402**.

(46) Refer to FIG. 1 and FIG. 6 together. FIG. 6 is a schematic structural diagram of a cushion fastener in FIG. 4 according to another embodiment. Generally, two tabs **12** are provided, and two corresponding second accommodating sub-grooves **2402** are provided. For example, when the two tabs **12** are disposed on a same side of the wound body portion **10**, the two second accommodating sub-grooves **2402** are disposed on a same side of the first accommodating sub-groove **2400**, and the two second accommodating sub-grooves **2402** may not communicate with each other. Further, when sizes of the two tabs **12** are different, sizes of the two corresponding second accommodating sub-grooves **2402** are also different. Therefore, it is necessary to ensure that an alignment relationship between the two tabs **12** and the two corresponding second accommodating sub-grooves **2402** is correct during placement. In order to reduce a difficulty of alignment during the placement, alignment marks may be provided at positions of the second accommodating sub-grooves **2402**.

(47) Certainly, this may also be implemented in another manner. For example, refer to FIG. 4 and FIG. 5. The two second accommodating sub-grooves **2402** in FIG. 6 may communicate with each other, that is, in this case, it is considered that one second accommodating sub-groove **2402** is present and configured to accommodate the two tabs **12**. Such design reduces the difficulty of alignment during placement.

(48) Still refer to FIG. 4, FIG. 5, and FIG. 6. In the second direction **L2**, one pick-and-place slot **248** is disposed on either of two opposite sides of the first accommodating sub-groove **2400**, and the pick-and-place slot **248** communicates with the first accommodating sub-groove **2400**. Use of the pick-and-place slot **248** can reduce difficulty of picking up or placing the wound battery cell **1**, and reduce the damage to the wound battery cell **1**. The second direction **L2** is perpendicular to the third direction **L3** in which the first accommodating sub-groove **2400** points to the second accommodating sub-groove **2402**, and in the third direction **L3**, a size of the first accommodating sub-groove **2400** is larger than a size of the pick-and-place slot **248**. For example, in the third direction **L3**, the size of the first accommodating sub-groove **2400** is twice the size of the pick-and-place slot **248**. Such design can reduce the impact on a position of the wound battery cell from the

pick-and-place slot **248** in use, to ensure the position limiting effect of the accommodating groove **240** on the wound battery cell.

(49) Still refer to FIG. **4**. The cushion fastener **244** is provided with a first error-proofing portion **241**, a second error-proofing portion **243** is disposed in the accommodating cavity **246**, and the first error-proofing portion **241** and the second error-proofing portion **243** cooperate with each other. Such design of the first error-proofing portion **241** and the second error-proofing portion **243** can reduce the probability of placing the cushion fastener **244** in the bearing member **242** in a wrong direction, improving inspection efficiency.

(50) Optionally, as shown in FIG. **4**, the cushion fastener **244** includes a plurality of corners, one of the corners is provided with a notch to form the first error-proofing portion **241**, and the corresponding second error-proofing portion **243** is a protrusion that matches the notch. Alternatively, in another embodiment, a specific area of the cushion fastener **244** is provided with a via hole, and a position of the accommodating cavity **246** corresponding to the via hole is provided with a protrusion.

(51) In addition, structures of the accommodating groove **240** in FIG. **4**, FIG. **5**, and FIG. **6** are designed to correspond to the square wound battery cell in FIG. **1**. When an object under test is a cylindrical wound battery cell in FIG. **3**, as shown in FIG. **7**, FIG. **7** is a schematic structural diagram of a cushion fastener in FIG. **4** according to another embodiment. The structure of the accommodating groove **240** on the cushion fastener **244** can be designed to be adapted to the cylindrical wound battery cell shown in FIG. **3**.

(52) Further, to distinguish the cushion fasteners **244** corresponding to wound battery cells of different models, as shown in FIG. **5**, FIG. **6**, and FIG. **7**, a surface of a side of the cushion fastener **244** facing away from the base plate **2420** may be provided with a marking area (that is, the position marked as A1 in FIG. **5**, FIG. **6**, and FIG. **7**). A type and a number of a battery cell under test are provided in the marking area, which helps quickly identify the cushion fastener **244** corresponding to the current wound battery cell under test.

(53) The gap inspection apparatus provided in this application is further described below from a perspective of a gap inspection method. Refer to FIG. **1**, FIG. **4**, and FIG. **8** together. FIG. **8** is a schematic flowchart of a gap inspection method according to an embodiment of this application. The gap inspection method includes the following steps.

(54) **S101**: Place a wound battery cell **1** in an accommodating groove **240** of a bearing mechanism **24**, and dispose the bearing mechanism **24** between an X-ray source **20** and an X-ray detector **22** that are oppositely disposed, where in a height direction P1 of the wound battery cell **1**, the wound battery cell **1** includes a wound body portion **10** and a tab **12** that are connected to each other, and the height direction P1 of the wound battery cell **1** crosses a first direction L1 in which the X-ray source **20** points to the X-ray detector **22**.

(55) Specifically, the wound battery cell **1** may be picked from a production line and put into the bearing mechanism **24**. In this case, the wound battery cell **1** is a bare cell, that is, in an unpackaged state. Generally, the bearing mechanism **24** is provided with a plurality of accommodating grooves **240**. In this case, one wound battery cell **1** may be placed in the accommodating grooves **240**, and inspection may be performed on each wound battery cell **1**.

(56) **S102**: Make the X-ray source **20** irradiate an arc-shaped area **106** of the wound body portion **10**, so that the X-ray detector **22** obtains an imaging image of the arc-shaped area **106**.

(57) Specifically, an X-ray emitted by the X-ray source **20** can penetrate the arc-shaped area **106** of the wound battery cell **1**, and the X-ray detector **22** converts a received X-ray into an electrical signal for imaging. Due to different penetrability of the X-ray for different structures and materials in a sample, an internal structure of the sample can be captured by the X-ray detector **22** and presented on a computer.

(58) **S103**: Obtain, based on the imaging image, a gap value between the two electrode plates within the arc-shaped area **106**.

(59) Specifically, refer to FIG. 9. FIG. 9 is a schematic flowchart of step S103 in FIG. 8

corresponding to an embodiment. A specific implementation process of step S103 includes:

(60) **S201**: Pre-process the imaging image and obtain a target area from the pre-processed imaging image.

(61) Specifically, a pre-processing process may include four steps: noise reduction by multi-frame superimposing and averaging, region of interest (ROI) interception, contrast enhancement, and filtering. Noise reduction by multi-frame superimposing and averaging is to superimpose and then average a plurality of frames of imaging images continuously collected by the X-ray detector 22, assign an average value to a corresponding new image, save a new image after noise reduction, and release memory, thereby reducing image noise and facilitating subsequent processing. ROI interception is to perform target area cropping on the new image to crop out a target area for inspection, to improve image processing accuracy and processing time. Contrast enhancement is to extract contrast of each part of an original image, highlight positive and negative electrode plate lines in the target area, and weaken an area not for inspection, so as to display the image more clearly. Filtering is to use a template to scan each pixel in the image and replace a value of a pixel point at a center of the template with a weighted average gray value of pixels in a neighboring area determined by the template, to make the image favorable for manual recognition. Specifically, as shown in FIG. 10, FIG. 10 is a schematic diagram of a target area obtained after step S201 in FIG. 9 according to an embodiment.

(62) **S202**: Determine position information of the positive electrode plate 100 and the negative electrode plate 102 from the target area based on size information of the positive electrode plate 100 and the negative electrode plate 102.

(63) Specifically, due to difference between the materials of the positive electrode plate 100 and the negative electrode plate 102, a width of the negative electrode plate 102 is wider than a width of the positive electrode plate 100. Therefore, positions of the positive electrode plates 100 and the negative electrode plates 102 may be determined from the target area based on the size information of the positive electrode plate 100 and the negative electrode plate 102.

(64) **S203**: Obtain, from the target area based on the position information of the positive electrode plate 100 and the negative electrode plate 102, a gap value between the positive electrode plate 100 and the negative electrode plate 102 that are adjacent to each other.

(65) Specifically, as shown in FIG. 10, after the positions of the positive electrode plates 100 and the negative electrode plates 102 are determined, a line may be drawn at a position 0.8 mm below a starting point of the negative electrode plate 102, and a center of the line is the brightest position of the negative electrode plate 102 (that is, a center position of the electrode plate). Similarly, a line may be drawn at a position 1 mm below a starting point of the positive electrode plate 100, and a center of the line is the brightest position of the positive electrode plate 100 (that is, a center position of the electrode plate). In this embodiment, for the convenience of distinction, a height of drawing the line of the positive electrode plate 100 is lower than a height of drawing the line of the negative electrode plate 102. Certainly, in another embodiment, another manner may also be used for distinction. For example, distinction is implemented by setting different line types and/or line colors, where the line types include solid line, dashed line, and the like. Subsequently, a gap value from a negative electrode line to a positive electrode line may be automatically measured, and the measured gap value is a distance from a center of the negative electrode plate 102 to a center of the positive electrode plate 100. Then, the measured gap value may be compared with a risk threshold, and if the gap value is greater than or equal to the risk threshold, it is determined that the wound battery cell is unqualified; or if the gap value is less than the risk threshold, it is determined that the wound battery cell is qualified. Inspectors can classify qualified and unqualified wound battery cells, and feedback an inspection result to a production line in a timely manner.

(66) To verify accuracy of the gap inspection method provided in this application, three wound battery cells were randomly selected from the production line, and the three wound battery cells

were tested by using the gap inspection method provided in this application and the CT slice inspection method in the prior art. Relative errors between gap values obtained by using the two testing methods were compared. As shown in table 1 below, table 1 lists comparison of test results obtained in the prior art and this application.

(67) TABLE-US-00001 TABLE 1 Comparison of test results obtained in the prior art and this application

	Relative error (last circle)	Relative error (last but one circle)	Test point 1	Test point 2	Test point 3	Test point 4	Test point 1	Test point 2	Test point 3	Test point 4	Sample 1	Sample 2	Sample 3
Test point 1	0.0%	8.7%	6.9%	6.9%	13.8%	17.5%	0.0%	17.8%	10.0%	3.6%	19.3%	17.6%	26.7%
Test point 2	10.7%	3.8%	16.7%	18.6%	11.1%	18.7%	13.8%	16.7%	11.1%	3.8%	16.7%		

(68) Specifically, relative error=|(measured value obtained by using CT slice inspection method-measured value obtained by using inspection method in this application)|/measured value obtained by using inspection method in this application*100%. As shown in Table 1, the relative errors between the gap values obtained by using the two methods were less than 20%, and the accuracy of the gap inspection method provided in this application is higher.

(69) In a specific application scenario, the gap inspection apparatus provided in this application uses a 150 KV micro-focus X-ray source, a flat panel detector with a pixel size less than or equal to 100 microns and a magnification greater than or equal to 5×, and a bearing mechanism with a bearing capacity of more than 25 KG. The bearing mechanism includes the bearing member and the cushion fastener. The base plate in the bearing member is made of a carbon fiber plate, the side plate is made of stainless steel, and a height of the accommodating cavity formed by the side plate and the base plate in the bearing mechanism in the first direction is 10 mm. The cushion fastener is made of a black EVA foam with a Shore hardness of 45 degrees and a height of 20 mm in the first direction. One of the corners of the cushion fastener has a triangular notch to form the first error-proofing portion, and the second error-proofing portion in a form of a triangular protrusion matching a shape of the triangular notch is disposed in the corresponding accommodating cavity. When the cushion fastener is disposed in the accommodating cavity, the first error-proofing portion and the second error-proofing portion abut against each other. In addition, a position of the cushion fastener adjacent to the first error-proofing portion is provided with a number, for example, a type and a number of a battery cell under test.

(70) The foregoing descriptions are merely the embodiments of this application, but are not intended to limit the patent scope of this application. Any equivalent structural transformations or equivalent process transformations made by using the content of the specification and drawings or direct or indirect application of the content of the specification and drawings to other related technical fields should fall within the patent protection scope of this application.

Claims

1. A gap inspection apparatus, characterized by being configured for a wound battery cell, wherein in a height direction (P1) of the wound battery cell, the wound battery cell comprises a wound body portion and a tab that are connected to each other, and the gap inspection apparatus comprises: an X-ray source and an X-ray detector that are oppositely disposed, and a bearing mechanism located between the X-ray source and the X-ray detector; wherein the bearing mechanism comprises at least one accommodating groove, the accommodating groove is configured to accommodate the wound battery cell, and the height direction of the wound battery cell crosses a first direction (LI) in which the X-ray source points to the X-ray detector; and the X-ray source is configured to irradiate an arc-shaped area of the wound body portion, and the X-ray detector is configured to obtain an imaging image of the irradiated arc-shaped area, so as to obtain a gap value between two electrode plates within the arc-shaped area based on the imaging image; wherein the bearing mechanism comprises a bearing member and a cushion fastener, an accommodating cavity is comprised on a side of the bearing member facing the X-ray source or the X-ray detector, the

cushion fastener is located in the accommodating cavity, and the accommodating groove is disposed on the cushion fastener; characterized in that: the accommodating groove comprises a first accommodating sub-groove and a second accommodating sub-groove that communicate with each other, the first accommodating sub-groove is configured to accommodate the wound body portion, and the second accommodating sub-groove is configured to accommodate the tab, wherein a preset distance is present between the wound body portion and an inner wall of the first accommodating sub-groove, and a preset distance is present between the tab and an inner wall of the second accommodating sub-groove.

2. The gap inspection apparatus according to claim 1, wherein: in the first direction (L1), the accommodating groove runs through the cushion fastener; and/or the cushion fastener comprises foam.

3. The gap inspection apparatus according to claim 1, wherein: two tabs are provided and arranged on a same side of the wound body portion; and the second accommodating sub-groove is configured to accommodate the two tabs.

4. The gap inspection apparatus according to claim 1, wherein: in a second direction (L2), one pick-and-place slot communicating with the first accommodating sub-groove is disposed on either of two opposite sides of the first accommodating sub-groove, wherein the second direction (L2) is perpendicular to a third direction (L3) in which the first accommodating sub-groove points to the second accommodating sub-groove, and a size of the accommodating groove in the third direction (L3) is larger than that of the pick-and-place slot.

5. The gap inspection apparatus according to claim 1, wherein the cushion fastener is provided with a first error-proofing portion, a second error-proofing portion is disposed in the accommodating cavity, and the first error-proofing portion and the second error-proofing portion cooperate with each other.

6. The gap inspection apparatus according to claim 1, wherein the bearing member comprises a base plate and a side plate that are connected to each other, and the side plate is arranged around the base plate to form the accommodating cavity; wherein a material of the base plate comprises carbon fiber, and a material of the side plate comprises metal.

7. The gap inspection apparatus according to claim 6, wherein the bearing member further comprises two handles that are oppositely disposed and located on an outer surface of a side of the side plate facing away from the accommodating cavity.

8. A gap inspection method, comprising using the gap inspection apparatus according to claim 1, wherein the gap inspection method comprises: placing a wound battery cell in an accommodating groove of the bearing mechanism, and disposing the bearing mechanism between an X-ray source and an X-ray detector that are oppositely disposed, wherein in a height direction of the wound battery cell, the wound battery cell comprises a wound body portion and a tab that are connected to each other, and the height direction of the wound battery cell crosses a first direction in which the X-ray source points to the X-ray detector; making the X-ray source irradiate an arc-shaped area of the wound body portion, so that the X-ray detector obtains an imaging image of the arc-shaped area; and obtaining, based on the imaging image, a gap value between two electrode plates within the arc-shaped area.

9. The gap inspection method according to claim 8, wherein the wound body portion comprises a positive electrode plate and a negative electrode plate that are alternately arranged; and the step of obtaining, based on the imaging image, a gap value between two electrode plates in the wound body portion within the arc-shaped area comprises: pre-processing the imaging image and obtaining a target area from the pre-processed imaging image; determining position information of the positive electrode plate and the negative electrode plate from the target area based on size information of the positive electrode plate and the negative electrode plate; and obtaining, from the target area based on the position information of the positive electrode plate and the negative

electrode plate, a gap value between the positive electrode plate and the negative electrode plate that are adjacent to each other.
